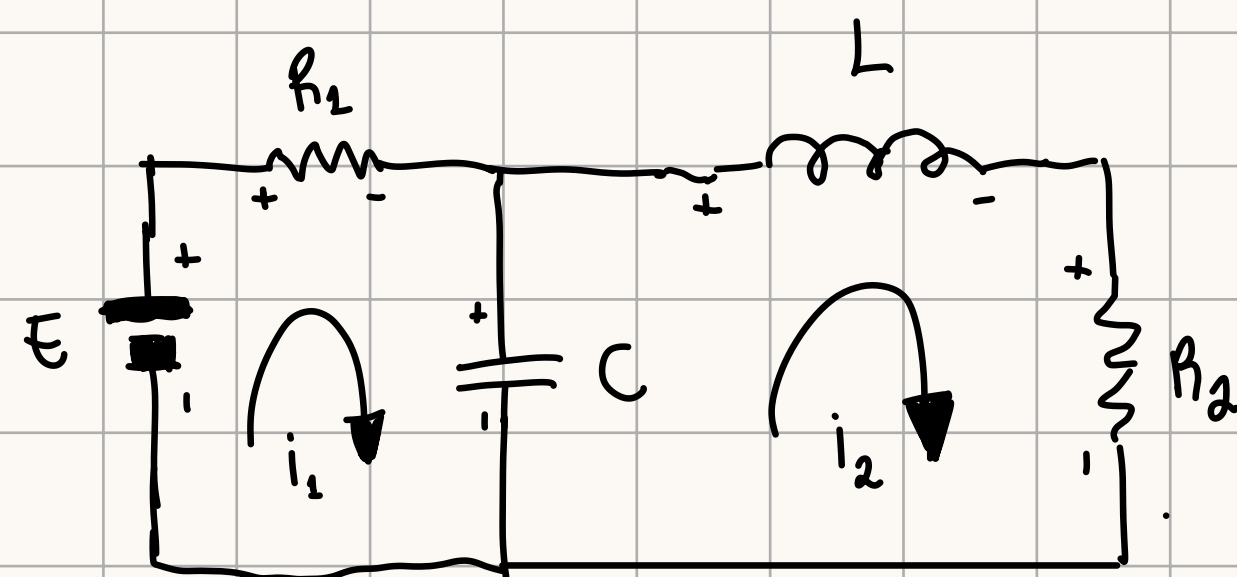


Sistemas dinámicos - Laboratorio N°3

Ejercicio # 11

$$L = 0.5H, C = 10\mu F \quad E(t) = 10V \text{ Salida } R_2(t)$$

$$R_1 = 30\Omega, R_2 = 20\Omega$$



Antes de resolver por mallas expresamos las impedancias en el dominio de la frecuencia para que sea más fácil trabajar

$$Z_{R1} = R_1, Z_{R2} = R_2$$

$$E(s) = \frac{10}{s}$$

$$Z_C = \frac{1}{sC}, Z_L = sL$$

$$Z_1 = Z_{R2} + Z_L = R_2 + sL = 20 + s0.5$$

$$Z_2 = Z_1 \parallel Z_C = \frac{(20 + s0.5) \cdot \frac{100}{s}}{20 + s0.5 + \frac{100}{s}}$$

$$= \frac{2000 + s50}{\left(20s + s^2 0.5 + 100\right)} \cdot \frac{s}{s} = \frac{2000s + s^2 50}{20s + s^2 + 100}$$

Usando KVL

$$I_2 = \frac{E - V_C}{R_1} \quad V_C = \frac{1}{s} \left(\frac{I_1 - I_2}{C} \right)$$

$$-E(s) + R_1 I_1 + Z_2 I_1 = 0$$

$$E(s) = I_1 (R_1 + Z_2)$$

$$I_1 = \frac{E(s)}{R_1 + Z_2}$$

$$V_{Z2} = \frac{E(s)}{R_1 + Z_2} \cdot Z_2$$

$$Z_2 = \frac{Z_C Z_1}{Z_C + Z_1}$$

$$I_2 = \frac{V_{Z2}}{Z_1} = \frac{E(s) \cdot Z_2}{(R_1 + Z_2) \cdot Z_1} = \frac{E(s) Z_C}{R_1 (Z_C + Z_1) + Z_C Z_1}$$

$$\frac{I_2}{E(s)} = \frac{Z_C}{R_1 (Z_C + Z_1) + Z_C Z_1}$$

$$H(s) = \frac{I_2(s)}{E(s)} = \frac{\frac{100}{s}}{30\left(\frac{100}{s} + s \cdot 0.5 + 20\right) + \frac{100}{s}(s \cdot 0.5 + 20)} \cdot \frac{s}{s}$$

$$= \frac{100}{\left(\frac{3000}{s} + s15 + 500 + 50 + 2000\right)} = \frac{100}{3000 + s^2 15 + s600 + s50 + 2000}$$

$$= \frac{100}{s^2(15) + s(650) + 5000}$$

$$E(s) = \frac{10}{s}, I_2(s) = E(s) \cdot H(s)$$

$$I_2(s) = \frac{1000}{s(s^2 15 + s(650) + 5000)} = \frac{200}{s(s)(3s+100)(s+20)} =$$

$$I_2(s) = \frac{200}{s(3s+100)(s+20)} = \frac{A}{s} + \frac{B}{s+20} + \frac{C}{3s+100}$$

Halum A

Halum B

$$\frac{200}{(s+20)(3s+100)} \Big|_{s=0} = \frac{1}{5} = A // \frac{200}{s(3s+100)} \Big|_{s=-10} = \frac{-2}{7} = B //$$

Halum C

$$\frac{200}{s(s+20)} \Big|_{s=-\frac{100}{3}} = \frac{9}{35} = C //$$

$$I_2(s) = \frac{1}{5s} + \frac{2}{7(s+20)} + \frac{9}{35(3s+100)}$$

$$= \frac{1}{5s} + \frac{2}{7(s+20)} + \frac{s}{35} \cdot \frac{1}{s+\frac{100}{3}}$$

$$\mathcal{L}^{-1}\{I_2(s)\} = \mathcal{L}^{-1}\left\{ \frac{1}{5s} - \frac{2}{7} \cdot \frac{1}{(s+10)} + \frac{3}{35} \cdot \frac{1}{\left(s + \frac{100}{3}\right)} \right\}$$

$$I_2(t) = \frac{1}{5} - \frac{2}{7} e^{-10t} + \frac{3}{35} e^{-\frac{100}{3}t} //$$